PVT-MAAG: A Hybrid Vision Transformer and Multi-Stage Adaptive Framework for Solar Photovoltaic Cell Defect Detection

Abstract

Nowadays, the usage of solar panels has become very popular. Photovoltaic (PV) cell defects can decrease the efficiency of these panels and can also become worse if these defects remain undetected. In this work, we have proposed an automated defect detection pipeline for electroluminescence images of PV cells. We have built this system on the modified version of the PVEL-AD dataset. This contains a total of 6,314 image samples across nine defect classes, and 2 background classes, the good class and the material class. These 2 classes do not have any annotation boxes. We have trained and compared a total of six baseline models. Among them, three YOLOv11n, YOLOv8m, and ResNet50-RetinaNet are CNN-based, and the remaining three, YOLOS, RT-DETR, and PVTv2B0-RetinaNet are Transformer-based. We have carried the PVTv2B0-RetinaNet forward, as it is our best-performing model, and have applied all our novelties to it. After completing all the Novelties, we have built our hybrid model, which we call PVT-MAAG. In PVT-MAAG, we have added adaptive low-rank fine-tuning (AdaLoRA), Bidirectional Mamba feature extraction, adaptive spatial feature fusion (ASFF), and graph-based reasoning. After performing all these novelties, our hybrid model achieved mAP@50 of 90.86% and mAP@50–95 of 60.61% on the test split and surpassed most of the baseline models. We have also distilled our PVT-MAAG model with two YOLO models. The Student YOLOv11n model achieves 6.94 points of mAP@50–95 from distillation, and YOLOv8m achieves 0.40 points. We have also produced Grad-Cam++ explanations for the plain PVTv2B0-RetinaNet model. The achieved hit rate is 44.9% across the test split. We ensembled ResNet50-RetinaNet with a PVTv2-based AdaLoRA and Mamba model using SAM-2 for segmentation and a rule-based fault severity estimator. With this, we have extended our system from a raw defect detector to one that can inspect PV cells and assess the severity of their defects.

Index Terms

AdaLoRA, Adaptive Spatial Feature Fusion, Bidirectional Mamba Feature Extraction, Electrolumienscene Imaging, Fault Severity Estimation, Grad-Cam++, Graph-based Reasoning, Knowledge Distillation, Object detection, Photovoltaic defect detection, PVEL-AD dataset, SAM-2 segmentation.

I. Introduction

Solar photovoltaics (PV) technology has experienced a rapid worldwide growth due to the rising need for renewable power sources. Global solar PV installed capacity had reached an estimated 2.2 TW in 2024 [1]. Such development makes it all the more important to maintain the trustworthiness and operation of PV cells. Defects and degradations may alter the efficiency of PV systems. PV modules have been reported to degrade at a median rate of approximately 0.5% per year [2]. Cell-level defects will cause the PV module to function improperly, and the severity will only increase over time. Several defects can occur during the manufacturing or operation of solar PV cells, such as cracks and finger defects [3]. Fig. 1 shows representative defect samples from the PVEL-AD dataset [4]. Early detection of such defects is necessary to prevent continued deterioration and preserve stable performance at all times.

Electroluminescent (EL) imaging is one of the most widely used methods for detecting defects and degradation in PV cells and modules [5]. Spatial information obtained by EL imaging enables the detection of defect areas that are difficult to evaluate with standard visual methods. EL image interpretation still requires thorough analysis to identify the smallest of defects. Manual inspection of defects is time-consuming and highly dependent on the inspecting individual's expertise. Such limitations require an automated method for efficient defect recognition in PV cells. Deep learning and computer vision models are among such methods. They can help detect defects in PV cells and also localise their spatial positions.

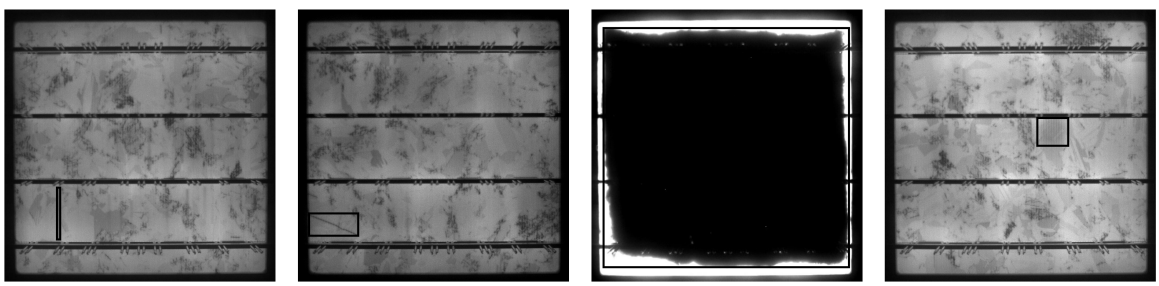


Fig. 1. Examples of four commonly observed PV cell defect classes in electroluminescence images, from left to right: finger, crack, black core, and thick line. Images retrieved from the PVEL-AD dataset [4]

Numerous recent works have investigated deep learning methods for automated defect detection in PV cells using EL images. These methods use convolutional neural networks (CNNs), object detection models, and transformer models to detect various types of PV cell defects.

Wang et al. [6] proposed MRA-YOLOv8 for better defect detection in PV cell EL images. They introduced a multi-branch coordinate attention network (MBCANet) into the backbone, a ResBlock module in the neck layer, and an alpha-minimum point distance-based IoU (AMPDIoU) loss function in the detection head. The authors evaluated their method on the PVEL-AD and SPDI datasets. They achieved mAP@50 of 91.7% on the PVEL-AD dataset and 69.3% on the SPDI dataset.

Mohamed et al. [7] proposed an improved and optimized version of the YOLOv12n model for defect detection in EL images of photovoltaic panels. In the proposed model they have replaced the C3k2 block at layer 20 with an A2C2f attention module, which comprises the Area Attention and Residual ELAN. They have tested the proposed model's performance on the PVEL-AD and Roboflow datasets. The model achieved the best mAP@50 of 91.30% on the PVEL-AD dataset and 90.90% on the Roboflow dataset, outperforming the baseline YOLOv12n model, which achieved the best mAP@50 of 90.20% on the PVEL-AD dataset and 88.20% on the Roboflow dataset.

Chen and his colleagues [8] tackled the problem of endogenous shift in photovoltaic defect detection, where images collected from different production lines or inspection heights vary in texture, brightness, and background style, causing existing detectors to perform poorly outside their training domain. They proposed the Gather and Distribute Domain shift Suppression Network (GDDS), a single-stage, single-domain generalization framework built around a DeepSpine backbone for multi-scale context capture and a Gather and Distribute module that fuses shallow and deep features to reduce background and instance-level shift. The authors also replaced the conventional CIOU loss with a normalized Wasserstein distance measure to make bounding box regression less sensitive to positional errors on small defects. Evaluated on an Electroluminescence Endogenous Shift Dataset and a Photovoltaic Inspection Infrared Dataset, GDDS achieved detection accuracies of up to 91.2% and 92.8% respectively, outperforming state-of-the-art two-stage, one-stage, and transformer-based detectors while also reducing computational complexity. The results show that GDDS generalizes noticeably better across unseen production lines and inspection heights than the baseline network, making it a practical option for real-world solar panel quality inspection.

Despite the excellent experiments, there still exist some drawbacks. The reviewed literature provides evidence of the limited application of contextual modelling, adaptive feature fusion, efficient fine-tuning, graph reasoning, knowledge distillation, and explainability. This gap was a key factor in the development of this proposed explanation-enhanced hybrid framework utilising Grad-CAM++.

The key contributions of this work can be summarised as follows:

- We introduce PVT-MAAG, a novel hybrid framework for PVTv2B0-RetinaNet, developed by combining Mamba-based contextual feature extraction, adaptive spatial feature fusion (ASFF), parameter-efficient fine-tuning with adaptive low-rank adaptation (AdaLoRA), and graph-based reasoning (GBR) to improve PV cell defect detection.
- Knowledge distillation was used with PVT-MAAG as the teacher model and YOLOv11n and YOLOv8m as student models, while SAM2-based segmentation and fault severity estimation were used to provide pixel-level defect illustration.
- The proposed hybrid system achieved better overall performance than standalone CNN- and transformer-based object detection models for detecting PV cell defects.

The remainder of this paper is organised as follows. Section II presents the proposed solar PV cell defect detection system. A thorough analysis of the investigated results is presented in Section III. Lastly, Section IV contains the concluding remarks and future work for this paper.

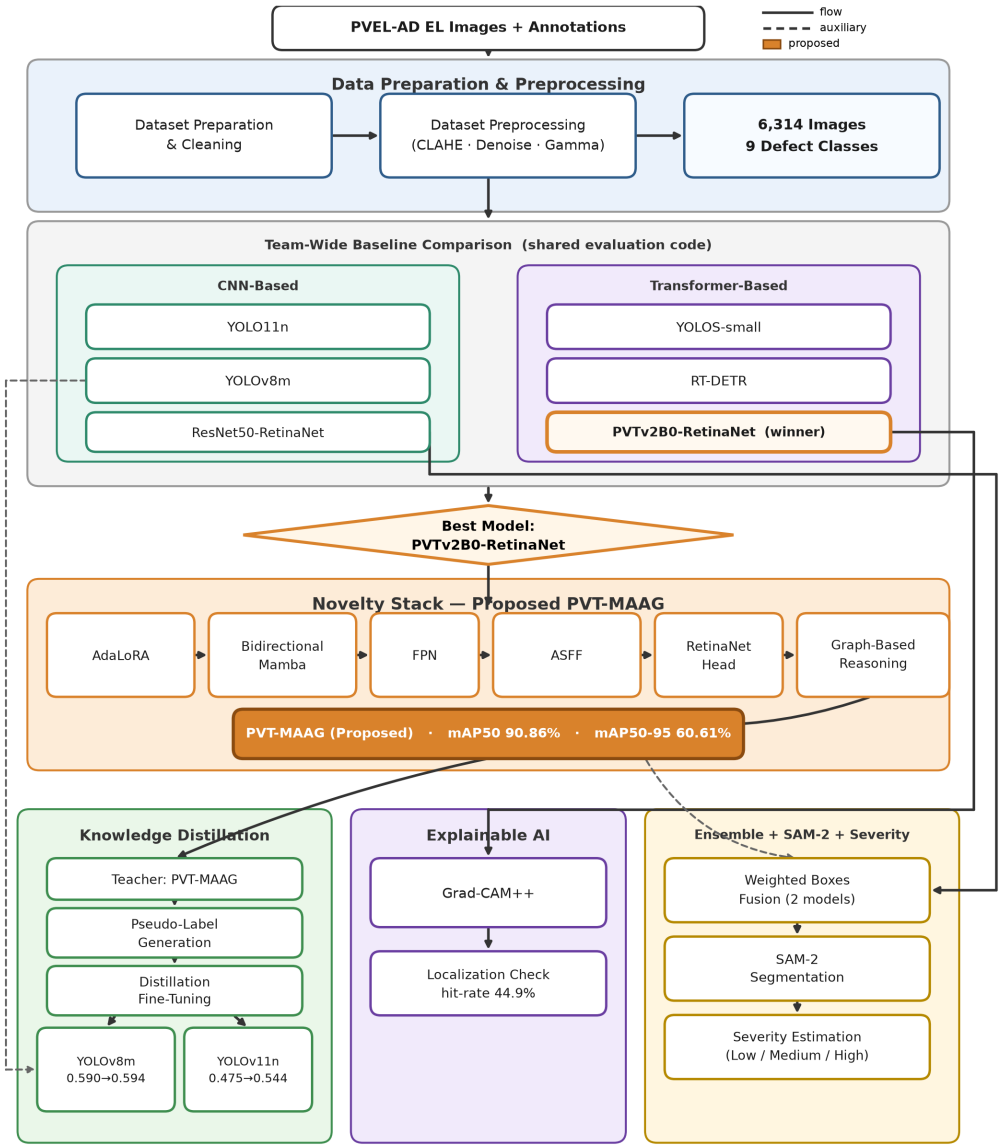


Fig. 2. Our Proposed System pipeline overview

In the Fig.2 we have show the our proposed system pipeline. From the dataset preparation, Preprocessing to the the implementation of our proposed novelties.From the Novelties to the final stage of the PVT-MAAG everything is showed here. Each stage is described in the proposed section II and the result is described in the section III.

II. Proposed System

A. Dataset

In our work, we have used the dataset of PVEL-AD (EL2025) [4]. It is a public dataset that contains the EL images of polycrystalline and monocrystalline PV cells.

`Dataset Link`: https://github.com/binyisu/PVEL-AD

In this dataset, the samples are distributed in `Pascal VOC` format. In this format, each of the image samples is paired with a XML file. This raw corpus contains the trainval split , a test split, and an othertypes pool. The othertype pool images samples contain the defect-free PV cell image samples of good cells and the material cells. None of these two classes contains any type of box annotation. We have processed the raw dataset and used it throughout our entire work. We have merged the trainval split, test split, and the good and material pool into one pool. Then we have removed the very rare classes. Any image samples with zero boxes after this, we have dropped them rather than keeping them as empty samples. After this, we have downsampled the remaining 9 defected classes to a target sample count agreed by all the members of the group. Separately, we have added 900 good-class image samples and 450 material-class image samples, which serve as background samples with no defect annotations. That is why we have used empty label files for these background image samples. After all these

processes, we have split the whole pool into a $70/15/15$ ratio and stratified the data by class with a fixed random seed. We have initialized the random seed to 42.

In the prepared dataset, we have total 6314 image samples. After the split, we have $4,418$ image samples in the train split, 947 image samples in the validation split, and 949 image samples in the test split. We have a total of 9 defective classes. Including `finger`, `crack`, `black core`, `thick line`, `horizontal dislocation`, `short circuit`, `vertical dislocation`, `star crack`, and `printing error`. This prepared dataset includes the two background classes the `good` and the `material` class.

TABLE I
CLASS-WISE DISTRIBUTION OF THE PROCESSED PVEL-AD DATASET

Class ID	Class Name	Train	Val	Test	Total
0	finger	786	177	187	1,150
1	crack	812	179	185	1,176
2	black_core	702	150	150	1,002
3	thick_line	697	149	149	995
4	horizontal_dislocation	555	119	119	793
5	short_circuit	525	112	113	750
6	vertical_dislocation	71	15	16	102
7	star_crack	139	30	30	199
8	printing_error	4	1	0	5
–	good (background)	630	135	135	900
–	material (background)	315	68	67	450
TOTAL IMAGE COUNT		**4,418**	**947**	**949**	**6,314**

Class IDs from 0 to 8 are label indices that we have used in this work to train every model. The `good` class and the `material` class serve as the background. They do not carry any detection classes and are stored in empty label files. That is why they do not have any class IDs.

B. Dataset Preprocessing

We have preprocessed our dataset to make the defects easier to learn. We have done all the preprocessing at the pixel level. We have not cropped, resized, or flipped any image samples during the preprocessing. We have kept all the box positions and label files unchanged. We have applied 4 steps in sequence, and each can be turned on or off independently.

- `Grayscale consistency`: EL sensors captured the image samples in the grayscale. But the source images store all three color channels with nearly equal values. We have collapsed the images into one channel and copied it back to remove the chroma noise introduced by compression.
- `Denoising`: To remove the sensor speckle noise before contrast is stretched, we have applied the denoising.
- `Contrast Limited Adaptive Histogram Equalization (CLAHE)`: CLAHE is a widely used technique in PV cell EL image defect detection. We have applied CLAHE to make the faint finger, crack, and dislocation more visible in the darker images.
- `Gamma Normalization`: We have applied gamma normalization in our prepared dataset. The exposure values vary across the captured batches and the gamma value for each image is automatically corrected. So, after applying the gamma normalization all the image brightness matches a common target value.

During preprocessing, we did not change the class distribution or the total sample count in our dataset. So our preprocessed dataset contains the same number of image samples that are already shown in the Class-wise Distribution of the Processed PVEL-AD Dataset table. We have applied the changes at the pixel level only.

C. Hyperparameter Tuning

Before the training, we have tuned our hyperparameters. To tune our hyperparameters we have used a generic algorithm search. This search started from a pretrained checkpoint and ran over our preprocessed dataset. In this hyperparameter tuning search, we have covered the learning rate schedule, momentum, weight decay, warmup length, detection loss weights, and the augmentation techniques, including the HSV jitter, geometric transforms, mosaic, mixup, cutmix, and copy paste probabilities.

We have run the search for 20 iterations. Each epoch is equivalent to 10 epochs. In total, equivalent to 200 epochs. It took around 5 and a half hours to complete on a single Nvidia RTX 4070 GPU. Every iteration trains a short run with one candidate hyperparameter set and scores them by validation mAP50-95. We have got our best hyperparameter at the iteration number 14. We have seen an improvement in this iteration of about three points over the first iteration score, which is 04468. Then we have saved these best hyperparameters and used it for our full training process.

TABLE II
TUNED HYPERPARAMETERS

Hyperparameter	Tuned Value
Initial learning rate	0.00967
Final learning rate factor	0.01229
Momentum	0.96583
Weight decay	0.00035
Warmup epochs	3.26089
Warmup momentum	0.66059
Box loss gain	6.26426
Class loss gain	0.58435
Distribution focal loss gain	1.54308

We have reused some of the hyperparameter values in knowledge Distillation while reconstructing a student model.

D. Applied Models

In this work, we have applied multiple CNN and transformer-based object detection models for the task of solar PV cell defect detection.

1) CNN-Based Object Detection Models:

- YOLOv11n: YOLOv11n is the nano version of YOLOv11, developed to offer effective object detection in terms of accuracy plus computing speed [9]. The model was selected as a lightweight model to compare with heavier ones.
- YOLOv8m: YOLOv8m is Ultralytic's medium sized an anchor free detection model [10]. Total parameter count of this model is 25.8 million. This is selected for comparision with the other heavy weight CNN based and Transformer based models model.
- ResNet50-RetinaNet: It is a torchvision based RetinaNet Model with the ResNet50 + FPN in the backbone. It is an anchor-based CNN model. Total parameter count for this model is around 31 million.

2) Transformer-Based Object Detection Models:

- PVTv2-B0–RetinaNet: PVTv2-B0 was used as the transformer-based backbone for RetinaNet. PVTv2 improves the original Pyramid Vision Transformer by implementing linear-complexity attention, overlap patch embedding, and convolutional feed-forward networks for dense vision tasks such as object detection [11].
- YOLOS: Our used YOLOS model is a pure ViT-based model without any CNN-based layer. We have the small version of the YOLOS model, and the total parameter count for this model is 30.65 million
- RT-DETR: RT-DETR: RT-DETR is a real-time transformer-based model combining a CNN-based backbone with a transformer-based encoder and Decoder. The total parameter count for our used RT-DETR model is around 20.08 million.

E. Mamba Feature Extraction

Mamba [12] is a selective state-space model that is designed for efficient sequence modeling. Mamba uses input-dependent state-space parameters to regulate the propagation of information through a sequence. Mamba's computing complexity is linear to the length of a sequence. In our work, we inserted a bidirectional Mamba between the PVTv2B0 backbone and the Feature Pyramid Network (FPN) module. The other features from the backbone layers were preserved without applying Mamba to them. We used the configuration $d_{\text{model}} = 256$, $d_{\text{state}} = 16$, $d_{\text{conv}} = 4$, with an expansion factor of 2.

As shown in Fig. 3, the deepest feature map $F \in \mathbb{R}^{B\times C\times H\times W}$ was flattened to form a sequence of length $L = H \times W$. It was subjected to layer normalization, then passed through the Mamba module. The normalized sequence was processed in both forward and reverse directions with the same set of Mamba parameters. In reverse processing, the order was first reversed prior to Mamba processing and then restored to the original order. The forward and reverse feature sequences are then fused as shown in (1).

$$Y_{\text{bi}} = \frac{Y_f + Y_r}{2} \tag{1}$$

where Y_f and Y_r are feature sequences processed in the forward and reverse passes through the Mamba module, respectively. Equation (2) shows the gated residual update. A learnable residual gate was introduced to control the contribution of the bidirectional feature sequences. It was initialized to 0.

$$X' = X + \tanh(g)Y_{\text{bi}} \tag{2}$$

where X is the base flattened feature sequence, and g is a learnable gate parameter.

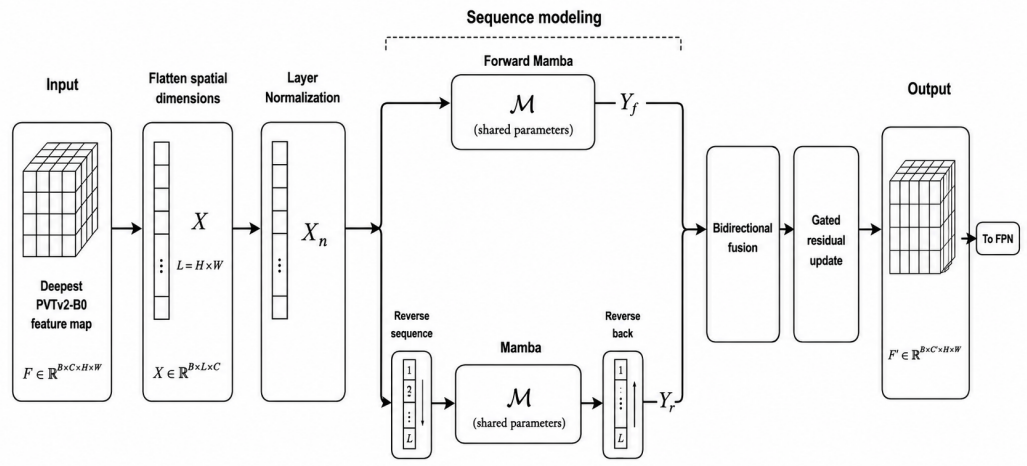


Fig. 3. Mamba-based feature extraction with bidirectional sequence processing in forward and reverse directions.

F. Adaptive Spatial Feature Fusion (ASFF)

ASFF [13] is a feature fusion method that adaptively fuses features extracted at different levels of the FPN based on their spatial relevance. Different feature locations are assigned spatially different weight values. This enables features at different scales to contribute differently to the resulting fusion. In our work, ASFF was applied after the FPN to the P3, P4, and P5 feature maps. P6 and P7 feature maps were preserved. We used compression channels $C_{\text{compress}} = 8$, fused levels $N_{\text{levels}} = 3$, softmax fusion normalization, and 3×3 refinement convolution. For each target pyramid level l, the fused feature vector at the spatial position (i, j) was calculated as depicted in (3).

$$Y_{ij}^{l} = \alpha_{ij}^{l} X_{ij}^{3\to l} + \beta_{ij}^{l} X_{ij}^{4\to l} + \gamma_{ij}^{l} X_{ij}^{5\to l} \tag{3}$$

where $X_{ij}^{3\to l}$, $X_{ij}^{4\to l}$, and $X_{ij}^{5\to l}$ represent the feature vectors at the spatial position (i, j) from the aligned feature maps of P3, P4, and P5, respectively. Their corresponding spatial weights are α_{ij}^{l}, β_{ij}^{l}, and γ_{ij}^{l}.

G. Adaptive Low-Rank Adaptation (AdaLoRA)

AdaLoRA [14] is a parameter-efficient fine-tuning technique in which the parameter budget of the low-rank space is adaptively allocated based on the importance of different weight matrices. It dynamically reallocates the low-rank budget during training by retaining important singular components and pruning less important ones. In our work, AdaLoRA was used to fine-tune the PVTv2-B0 backbone. The query projection q, combined key-value projection kv, and attention output projection $proj$ of the attention mechanism, along with the fully connected layers $fc1$ and $fc2$ of the MLP block, were adapted. We used initial rank $r_{\text{init}} = 12$, target rank $r_{\text{target}} = 8$, scaling factor $\alpha = 32$, smoothing coefficients $\beta_1 = 0.85$, and $\beta_2 = 0.85$. Equation (4) shows the incremental update of a pretrained weight matrix, which is parameterized using an SVD-based low-rank decomposition [14].

$$W = W^{(0)} + \Delta = W^{(0)} + P\Lambda Q \tag{4}$$

where $W^{(0)}$ is the initial frozen pretrained weight matrix. P and Q are the left and right singular vectors of the incremental update, respectively. Λ is a diagonal matrix that contains the corresponding singular values.

H. Graph-Based Reasoning

1) Static Graph Based Reasoning: Graph-Based Reasoning [15] is a technique by which models create relationships between different regions of a sample image using the Graph Neural Network. This allows a detector to refine its prediction using contextual information from the other related regions instead of scoring each location independently. In our paper, we have inserted and implemented the Graph-based Reasoning between the RetinaNet head and the loss/non-maximum suppression stage. For each of the sample images, the top K with the highest scoring anchor locations were selected as the graph nodes. We selected the graph nodes with each node's features studied directly from its corresponding Adaptive Spatial Feature Fusion level and spatial position. Edges were formed as the union of K-Nearest Neighbor (KNN) edges between the center of the anchor cox and the grid band edges connecting edges that lie on the same finger, the latter identified per image through peak detection over the row-wise and column-wise pixel intensity profiles. A two-layer, four-head graph attention network refined the node features and the correction head. Initialized to zero and mapped the refined feature to a per class logic delta. We

have used the top $K = 128$ nodes and $K = 6$ nearest neighbors. We have initialized the learnable gate parameter $\alpha = 0.1$. The corrections were added residually to the classification logits at the selected anchor places. We can see it in equation (5)

$$Z' = Z + \alpha \cdot C(H) \tag{5}$$

Here,

- $Z =$ Original classification logits
- $H =$ Graph Attention Refined node features
- $C(H) =$ Liner correction head
- $\alpha =$ Learnable gate parameter

2) Topology-Aware Dynamic Graph Neural Network : On our PVT-MAAG, we have applied the Topology-Aware Dynamic Graph Neural Network (TA-DGNN). We have placed the TA-DGNN in the middle of the RetinaNet head and the loss and Non-max suppression stage.

The same rule that we have used for the selection of nodes for static GBR, we have used. They read the top K highest scoring anchors per image and features from their ASFF and spatial location.

The TA-DGNN does not have any fixed edges. As shown in the equation 6, each layer re-aggregates the attended nodes with the current node features and the geometry, and passes the result through a small network B to bias the raw attention score.

$$A_{ij} = \frac{Q_i \cdot K_j}{\sqrt{d}} + B(g_{ij}) \tag{6}$$

We have embedded a structural bias directly into attentions. It is the same principle as Reinforced Structural Attention in [16] which adds a topological heuristic term to attention for link predictions . We kept only the K closest neighbors by this score. So neighbor list is different in each layer. In the Topological-Aware Dynamic Graph Neural Network, we have in total 4 heads and kept K=6. neighbors.

Here , each of the updates to the neighbors updates the nodes via the softmax-weighted value aggression , a residual connection and an FFN . which is a transformer layer. Our TA-DGNN consists of two stacked layers.

The refined features are passed through a correction head and residually added to a classification logits at selected anchors via a learnable gate. We have shown it in the equation 7.

$$Z' = Z + \tanh(g) \cdot C(H) \tag{7}$$

Here,

- $Z =$ Original classification logits
- $H =$ Refined node features
- $C =$ Correction head
- $g =$ Learnable gate parameter

I. Knowledge Distillation

Knowledge Distillation [17] is a model compression technique in which a smaller student model is trained to reproduce the behavior of an already-trained teacher model, transferring the teacher's learned knowledge into a form that is cheaper and easier to deploy. This technique has previously been used to obtain lightweight, deployable models for photovoltaic cell defect detection from larger pre-trained teacher networks [18]. In our work, the Graph-Based-Reasoning-enhanced detector (PVTv2-B0 + BiMamba + FPN + ASFF + AdaLoRA + RetinaNet + GBR) was used as a single frozen teacher, distilled into two independent students. One is the YOLOv8 model,, and the other is the YOLOv11n model, to cover both mid-size and very lightweight deployment targets. Here , the teacher model and the student model do not share any compatible output spaces. That is why we have used the response-based pseudo-label distillation rather than a direct logit or feature matching. We ran the training on the training sample images for the frozen teacher model once. Then, its detections above a confidence threshold were cached as pseudo-labels in YOLO format. We once fine-tuned both of the student models against the real labels and then again fine-tuned both of them using two loss passes per batch. We have used one loss pass against the real ground truth and another against the teacher models' pseudo-labels. We summed both before backward error propagation, as shown in equation (6). First bit of the student models. We have initialized the pseudo-label confidence threshold to 0.5, weighted factor $\lambda_{\mathrm{KD}} = 0.5$, image size to 640, and effective batch size to 16. For both of the models, we have kept the AdamW at a learning rate 10^{-4}, weight decay $5{\times}10^{-4}$. Trained for 15 epochs with the patience 5.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{real}} + \lambda_{\mathrm{KD}}\, \mathcal{L}_{\text{pseudo}} \tag{8}$$

Here,

- $\mathcal{L}_{\text{real}} =$ Detection loss is computed against the real Ground Truth labels
- $\mathcal{L}_{\text{pseudo}} =$ Detection loss computed against the teacher's pseudo-labels
- $\lambda_{\mathrm{KD}} =$ Fixed weighting hyperparameter.

J. Model Ensemble

In order to enhance performance, we combined a CNN-based RetinaNet (ResNet-50-FPN) and a transformer-based RetinaNet model. The transformer-based model includes PVTv2-B0, BiMamba, ASFF, and AdaLoRA. Weighted Boxes Fusion (WBF) [19] was used to implement prediction-level ensembling. The Non-Maximum Suppression (NMS) method eliminates overlapping boxes and selects only the best-scoring one, whereas WBF builds a new box by calculating a confidence-weighted average of all boxes that are similar. Therefore, WBF uses all boxes' information while NMS discards them.

To obtain the corresponding detection set, we used both detectors on each test image, producing predictions of the form (b_i, s_i, c_i), where b_i denotes the bounding box coordinates, s_i the confidence score, and c_i the predicted class. An Intersection over Union (IoU) threshold of 0.55 was used to pool and cluster the predictions of both models per class. For each cluster of N matched boxes, the fused box was computed as the confidence-weighted average shown in (9), and the fused confidence score was computed as the mean of the contributing scores, shown in (10).

$$b_{\text{fused}} = \frac{\sum_{i=1}^{N} s_i \, b_i}{\sum_{i=1}^{N} s_i} \tag{9}$$

$$s_{\text{fused}} = \frac{1}{N} \sum_{i=1}^{N} s_i \tag{10}$$

Equal fusion weights were used for both models ($w_1 = w_2 = 1$), and detections below a confidence threshold of 0.05 were discarded prior to fusion. The ensemble was evaluated against each individual model on the held-out test set using COCO-style mean Average Precision (mAP) at IoU thresholds of 0.50 and $0.50{:}0.95$, along with Precision, Recall, and F1-score computed at a fixed confidence threshold of 0.5 and IoU threshold of 0.5. The ensemble improved upon both individual models across every metric, most notably at the stricter IoU $= 0.75$ threshold and on small-object detection. This indicates that the two architectures exhibit complementary localization behavior rather than redundant predictions. As detailed in Section III-A, this ensemble configuration achieved the best overall performance among all evaluated configurations and was therefore selected as the final model for the fault severity estimation and SAM-2 segmentation stages described below.

K. SAM-2 Based Segmentation

To obtain pixel-level delineation of detected defects, we incorporated the Segment Anything Model 2 (SAM-2) [20]. It provides segmentation beyond the coarse bounding-box output of the detection stage. Since the dataset does not include pixel-level (mask) annotations, SAM-2 was applied directly using its pretrained weights without any dataset-specific fine-tuning.

For each ensemble-fused detection, the predicted bounding box was passed to SAM-2 as a box prompt. SAM-2's image encoder and mask decoder then produced a binary segmentation mask $M \in 0, 1^{H \times W}$ corresponding to the defect region within the prompted box, using its interactive segmentation formulation:

$$M = \text{SAM-2}(I, b) \tag{11}$$

where I is the input EL image and b is the detector-predicted bounding box used as the prompt. The lightweight Hiera-Small variant of SAM-2 was used to balance segmentation quality against inference cost across the full test set.

The resulting masks were used as an auxiliary signal for area-based analysis in the fault severity estimation stage described below. Specifically, the segmented defect area was explored as an alternative to bounding-box area for estimating defect severity. As no ground-truth segmentation masks are available for this dataset, the SAM-2 outputs are used qualitatively and as an auxiliary severity signal, rather than being validated quantitatively using mask-level metrics such as mean IoU or Dice score.

L. Fault Severity Estimation

As the dataset provides only class-level defect annotations with no severity labels or ground-truth grading, fault severity was estimated using a heuristic, data-driven scoring scheme rather than a supervised classifier. Severity was derived from two signals available at inference time: the spatial extent of the detected defect relative to the solar cell, and the inherent electrical/structural risk associated with the defect class.

1) Area-Based Severity: For each confident detection ($s \geq 0.50$, matching the threshold used for Precision/Recall/F1 evaluation), the fraction of the image area occupied by the predicted bounding box was computed as in (12).

$$r = \frac{w_{\text{box}} \times h_{\text{box}}}{W_{\text{img}} \times H_{\text{img}}} \tag{12}$$

Rather than fixing arbitrary severity thresholds, bucket boundaries were derived directly from the empirical distribution of r over the test set, using the 33rd and 66th percentiles to define approximately balanced Low, Medium, and High severity categories, as shown in (13).

$$\text{Severity}(r) = \begin{cases} \text{Low} & r < q_{33} \\ \text{Medium} & q_{33} \leq r < q_{66} \\ \text{High} & r \geq q_{66} \end{cases} \tag{13}$$

2) Class-Weighted Severity Score: Because raw area alone does not capture the differing real-world risk posed by different defect types (e.g., a *short_circuit* is inherently more critical to panel performance than a cosmetic *printing_error* of similar size), each class was assigned a fixed severity weight ω_c reflecting its relative electrical/structural impact, informed by domain characteristics of EL defect types. The final weighted severity score for a detection was computed as in (14).

$$S_{\text{weighted}} = r \times \omega_c \tag{14}$$

3) Limitation: Localization-Dependent Reliability: Analysis of the per-class severity distribution revealed that area-based severity is not equally reliable across all defect classes. Specifically, the *short_circuit* class exhibited a median bounding-box-to-image area ratio of 0.51 (maximum 0.90), substantially larger than well-localized classes such as *finger*, *crack*, and *thick_line* (all below 2% median). The *black_core* class showed a smaller but still elevated median ratio of 0.12. Since both *short_circuit* and *black_core* were minority classes requiring inverse-frequency weighted sampling during training (Table I), we attribute this to comparatively weaker box-regression precision for underrepresented classes rather than genuinely large physical defect extents. Consequently, area-based severity scores for these two classes should be interpreted with reduced confidence relative to the class-weighted component of the score, and this is reported as an identified limitation of the current severity estimation approach rather than corrected post hoc, to preserve the interpretability of the reported metric.

M. Explainable AI

To increase the transparency of our work, we have applied Explainable AI. As the XAI, we have used the Grand-Cam++. We have applied the XAI on our best performing plain PVTv2b0 + FPN + RetinaNet model. With this, we have produced the [21] visual heatmap for our best-performing PVTv2b0 + FPN + RetinaNet model. After applying the Explainable AI (XAI) our achieved overall hit rate is 44.9%.

III. Results and Discussion

This section presents the results obtained during the implementation of this work.

Table III presents the performance comparison of the applied CNN- and transformer-based object detection models for PV cell defect detection. The best-performing models for each performance metric are presented in bold. The proposed PVT-MAAG model attained the highest overall detection performance, except for precision and Dice score, where ResNet50-RetinaNet performed slightly better, whereas RT-DETR had a slightly higher IoU.

TABLE III
Comprehensive Performance Comparison of Various Models for PV Cell Defect Detection

Model	Precision	Recall	F1	IoU	Dice	mAP@50	mAP@50–95
YOLOv11n	0.7494	0.7880	0.7682	0.8005	0.8846	0.7922	0.4745
YOLOv8m	0.6409	0.8176	0.7133	0.7995	0.8845	0.9069	0.5904
ResNet50-RetinaNet	**0.8904**	0.7410	0.8089	0.8351	**0.9064**	0.8776	0.5976
PVTv2B0-RetinaNet	0.7366	0.8995	0.8100	0.8085	0.8898	0.8852	0.5626
YOLOS	0.5715	0.7791	0.6473	0.8111	0.8898	0.8215	0.5404
RT-DETR	0.5605	0.8975	0.6900	**0.9246**	0.8646	0.8292	0.5991
PVT-MAAG	0.7327	**0.9288**	**0.8191**	0.8268	0.9013	**0.9085**	**0.6065**

The complexity of the models and their training efficiency have been summarised in Table IV, which provides information on model size, number of parameters, and training time for each epoch. YOLOv11n was the most lightweight model, with 2.59M parameters, 5.22 MB in size, and only 1.07 minutes per epoch for training. YOLOS was the most complex model with 30.65M parameters and 116.96 MB in size. The proposed PVT-MAAG model had 14.23M parameters, showing a moderate increase in complexity over the 11.47M parameters of PVTv2B0-Retinanet.

Fig. 4 presents the normalised confusion matrix for the hybrid PVT-MAAG model on the test split for PV cell defect detection. Most classes have high accuracy on the diagonal. Classes such as horizontal_dislocation (1.00), black_core (0.98), short_circuit (0.98), and vertical_dislocation (0.98) have near-perfect accuracy. The common mistakes include missed detections that fall into the background column, such as crack (0.17), thick_line (0.14), and finger (0.12), as well as background detections that are classified as crack.

TABLE IV
COMPARISON OF APPLIED MODELS IN TERMS OF COMPLEXITY AND TRAINING EFFICIENCY.

Model	Size (MB)	Parameters	Training time per epoch (min)
YOLOv11n	5.22	2.59 M	1.07
YOLOv8m	49.63	25.84 M	1.40
ResNet50-RetinaNet	123.86	32.00 M	7.00
PVTv2B0-RetinaNet	43.82	11.47 M	3.66
YOLOS	116.96	30.65 M	4.15
RT-DETR	76.96	20.08 M	6.4
PVT-MAAG	53.77	14.23 M	4.98

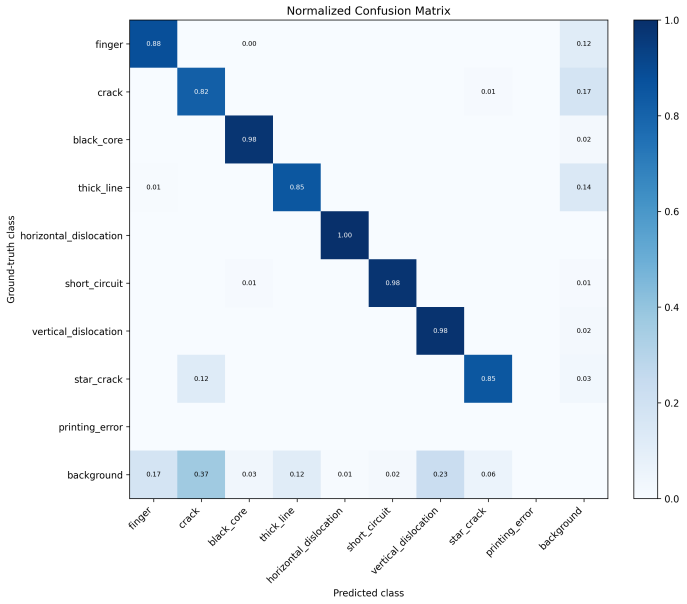


Fig. 4. Normalized confusion matrix for PVT-MAAG on the test split.

Fig. 5 shows the training and validation loss versus epoch curves. Training loss falls gradually from 0.58 to 0.07, and the maximum fall is in the first ten epochs, after which it becomes flatter. Validation loss follows the trend of training loss until epoch 10, then diverges, becomes fluctuating in the middle epochs, and flattens to 0.14-0.15 after epoch 30, whereas training loss continues to fall.

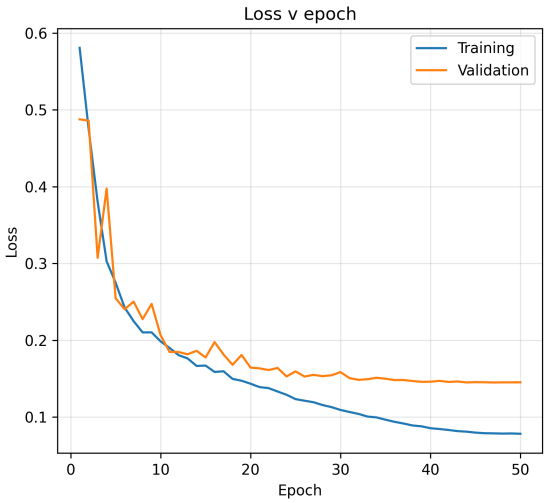


Fig. 5. Training and validation loss curves across 50 epochs.

Fig. 6 presents the training and validation mAP@50-95 versus epoch curves for the proposed PVT-MAAG hybrid model. The two curves increase together from about 0.12 to 0.55 over the first 20 epochs. After 20 epochs, training mAP@50-95 continues to increase, whereas validation mAP@50-95 remains constant at approximately 0.58-0.59 from epoch 25 onwards.

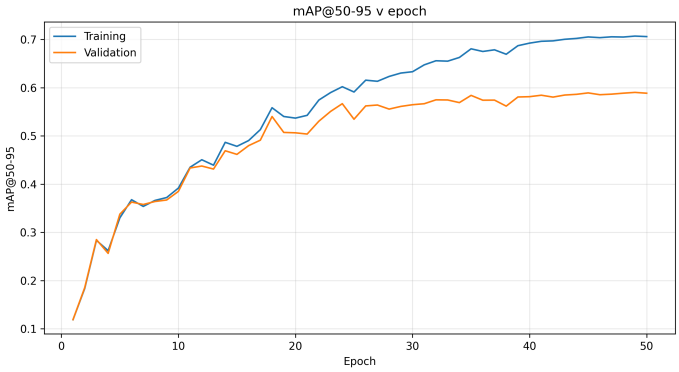


Fig. 6. Training and validation mAP@50-95 versus epoch.

Table V shows a comparison of cross-dataset generalisation of the hybrid PVT-MAAG model on an independent EL dataset. The dataset we used for this was the PV-Multi-Defect Dataset [22]. The model was tested on this dataset with and without fine-tuning. Without fine-tuning, performance is minimal in both cases, resulting in only 0.0053 mAP@50-95, with F1-scores of 0.0193 for Static GBR and 0.0192 for TA-DGNN. After fine-tuning on the independent dataset, performance improves substantially for both models. PVT-MAAG with static GBR achieves 0.5807 F1-score and 0.3552, and with TA-DGNN, 0.5902 F1-score and 0.3373 mAP@50-95 after fine-tuning.

TABLE V
CROSS-DATASET PERFORMANCE OF PVT-MAAG WITH STATIC GBR AND TA-DGNN ON THE INDEPENDENT PV-MULTI-DEFECT DATASET

PVT-MAAG Configuration	Fine-tuning	Precision	Recall	F1	IoU	Dice	mAP@50	mAP@50-95
Static GBR w/o Fine-tuning	×	0.0212	0.0176	0.0193	0.6980	0.8130	0.0079	0.0053
Static GBR w/ Fine-tuning	✓	0.4605	0.7860	0.5807	0.7737	0.8669	0.6813	0.3552
TA-DGNN w/o Fine-tuning	×	0.0211	0.0176	0.0192	0.6977	0.8219	0.0079	0.0053
TA-DGNN w/ Fine-tuning	✓	0.4806	0.7647	0.5902	0.7725	0.8717	0.6722	0.3373

A. Ablation Study

The ablation results show the performance improvement achieved through systematically adding every component to the baseline PVTv2B0-RetinaNet model.

As shown in Table VI, applying augmentation to baseline PVTv2B0-Retinanet increased mAP@50-95 to 0.5658, and Mamba increased mAP@50-95 to 0.5906. The inclusion of ASFF yields a 0.6046 mAP@50-95. Adding AdaLoRA gave the maximum mAP@50-95 and F1-score, at 0.6065 and 0.8151, respectively. The inclusion of GBR (TA-DGNN) results in a small increase in F1-score from 0.8151 to 0.8191. Overall, the full PVT-MAAG framework improved mAP@50-95 by 0.0439 over the baseline.

TABLE VI
CUMULATIVE ABLATION STUDY OF PROPOSED PVT-MAAG FRAMEWORK EVALUATING THE CONTRIBUTION OF EACH COMPONENT

Configuration	**Aug.**	**Mamba**	**ASFF**	**AdaLoRA**	**GBR**	**F1**	**mAP@50**	**mAP@50–95**
PVTv2B0-RetinaNet (Baseline)	×	×	×	×	×	0.8100	0.8852	0.5626
+ Aug.	✓	×	×	×	×	0.7964	0.8939	0.5658
+ Aug. + Mamba	✓	✓	×	×	×	0.8102	0.9036	0.5906
+ Aug. + Mamba + ASFF	✓	✓	✓	×	×	0.8133	0.9073	0.6046
+ Aug. + Mamba + ASFF + AdaLoRA	✓	✓	✓	✓	×	0.8151	0.9087	0.6065
+ Aug. + Mamba + ASFF + AdaLoRA + GBR	✓	✓	✓	✓	✓	0.8191	0.9085	0.6065

Table VII demonstrates the performance of PV cell defect detection according to the scan direction of Mamba scans. The performance of the augmented baseline has been enhanced by both Mamba scan directions. Bidirectional Mamba has performed best, with mAP@50 and mAP@50-95 metrics of 0.9066 and 0.5986, respectively.

Comparison results for the hybrid PVT-MAAG model without graph reasoning, with static GBR, and with TA-DGNN are presented in Table VIII. The hybrid PVT-MAAG model without graph reasoning and with TAD-GNN attained the highest

TABLE VII
COMPARISON OF UNIDIRECTIONAL AND BIDIRECTIONAL MAMBA FOR PV CELL DEFECT DETECTION

Configuration	F1	mAP@50	mAP@50–95
PVTv2B0-RetinaNet + Aug.	0.7964	0.8939	0.5658
+ Unidirectional Mamba	0.8092	0.9014	0.5862
+ Bidirectional Mamba	**0.8102**	**0.9036**	**0.5906**

mAP@50-95 of 0.6065. The PVT-MAAG model with TA-DGNN achieved a 0.7327 precision, 0.9288 recall, 0.8191 F1-score, 0.8268 IoU, 0.9013 Dice coefficient, 0.9085 mAP@50, and 0.6065 mAP@50-95. TA-DGNN improved F1-score marginally from 0.8148 to 0.8191 and mAP@50-95 from 0.6061 to 0.6065 compared to static GBR.

TABLE VIII
PERFORMANCE COMPARISON OF STATIC AND DYNAMIC GRAPH REASONING IN PVT-MAAG

Graph Reasoning	Precision	Recall	F1	IoU	Dice	mAP@50	mAP@50–95
None	0.7262	0.9288	0.8151	0.8271	0.9015	0.9087	0.6065
Static GBR	0.7258	0.9288	0.8148	0.8269	0.9013	0.9086	0.6061
TA-DGNN	**0.7327**	**0.9288**	**0.8191**	0.8268	0.9013	0.9085	**0.6065**

B. Knowledge Distillation

We have applied the knowledge distillation in our work here. As the Teacher model, we have used the PVT-MAAG + GBR, and as the student model, we applied the YOLOv8m model and the YOLOv11n model. Both student models were evaluated on the test split and reported both the results and performance before and after applying the distillation by keeping the teacher model the same.

TABLE IX
KNOWLEDGE DISTILLATION RESULTS COMPARED TO TEACHER AND ORIGINAL MODELS

Model	Preciosion	Recall	F1	mAP@50	mAP@50–95	Params	ΔmAP50-95 vs original
Teacher (PVT-MAAG + GBR)	0.7258	0.9288	0.8148	0.9086	0.6061	14.06 M	–
YOLOv8m (original)	0.8359	0.8754	0.8552	0.9069	0.5904	25.84 M	–
YOLOv8m (KD-distilled)	0.8017	0.8921	0.8445	**0.9094**	**0.5944**	25.84 M	**+0.0040**
YOLOv11n (original)	0.7495	0.7882	0.7684	0.7923	0.4747	2.58 M	–
YOLOv11n (KD-distilled)	**0.8004**	**0.8166**	**0.8084**	**0.8628**	**0.5441**	2.58 M	**+0.0694**

Table IX demonstrates the performance of PV cell defect detection of our distilled student models before and after performing the knowledge Distillation

C. Model Ensemble

To further improve the robustness of the detection framework, a model ensemble was constructed. By combining the predictions of PVT-MAAG and ResNet50-RetinaNet using Weighted Boxes Fusion (WBF). An ensemble ablation study was conducted to evaluate whether combining the two complementary detectors provides an advantage over using either model individually. Both models were evaluated independently on the test set, and their predictions were subsequently fused using WBF under the same evaluation process.

As shown in Table X, PVT-MAAG achieved a precision of 0.7258, recall of 0.9288, F1-score of 0.8148, mAP@50 of 0.9086, and mAP@50–95 of 0.6061. In comparison, ResNet50-RetinaNet achieved a substantially higher precision of 0.8904 but a lower recall of 0.7410, resulting in an F1-score of 0.8089, mAP@50 of 0.8776, and mAP@50–95 of 0.5976. This difference in performance characteristics indicates that the two detectors provide complementary predictions: PVT-MAAG is more effective at identifying a larger proportion of defects, on the other hand ResNet50-RetinaNet produces fewer false-positive detections.

When the predictions of both detectors were combined using WBF, the resulting ensemble achieved the best overall performance. Precision of 0.9257, F1-score of 0.8413, mAP@50 of 0.9140, and mAP@50–95 of 0.6348. Compared with PVT-MAAG alone, the ensemble improved precision from 0.7258 to 0.9257 and increased F1-score from 0.8148 to 0.8413. The mAP@50–95 also increased from 0.6061 to 0.6348, corresponding to an improvement of 0.0287. Compared with ResNet50-RetinaNet, the ensemble increased mAP@50–95 from 0.5976 to 0.6348. These improvements demonstrate that WBF effectively exploits the complementary predictions of the two detectors and produces more reliable and accurately localized defect predictions.

TABLE X
ENSEMBLE ABLATION STUDY EVALUATING INDIVIDUAL DETECTORS AND WEIGHTED BOXES FUSION (WBF)

Configuration	PVT-MAAG	ResNet50-RetinaNet	WBF	Precision	Recall	F1	mAP@50	mAP@50–95
PVT-MAAG	✓	×	×	0.7258	0.9288	0.8148	0.9086	0.6061
ResNet50-RetinaNet	×	✓	×	0.8904	0.7410	0.8089	0.8776	0.5976
PVT-MAAG + ResNet50-RetinaNet	✓	✓	✓	**0.9257**	0.7710	**0.8413**	**0.9140**	**0.6348**

D. SAM2-Based Defect Segmentation

To enhance defect localization, we used the detected bounding boxes as prompts for the Segment Anything Model 2 (SAM2). For each detection, the highest-scoring candidate mask was selected.

Fig. 7 shows the segmentation result for a *finger* defect. The detector achieves a confidence of 0.806. On the other hand, SAM2 produces a mask score of 0.861.

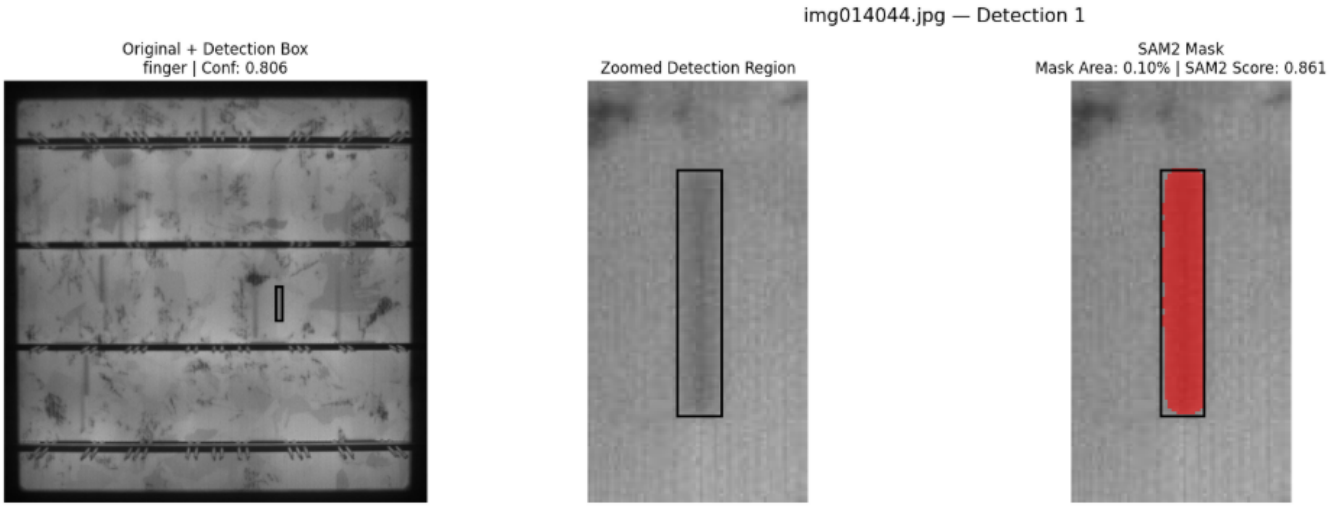


Fig. 7. SAM2-based segmentation of a finger defect.

Fig. 8 shows a *star-crack* defect with a detection confidence of 0.926. SAM2 obtains a mask score of 0.802. These examples show that the proposed detection-guided SAM2 approach can refine bounding-box detections into more detailed pixel-level representations. The reported SAM2 scores are predicted mask-quality scores and are not equivalent to IoU or Dice scores.

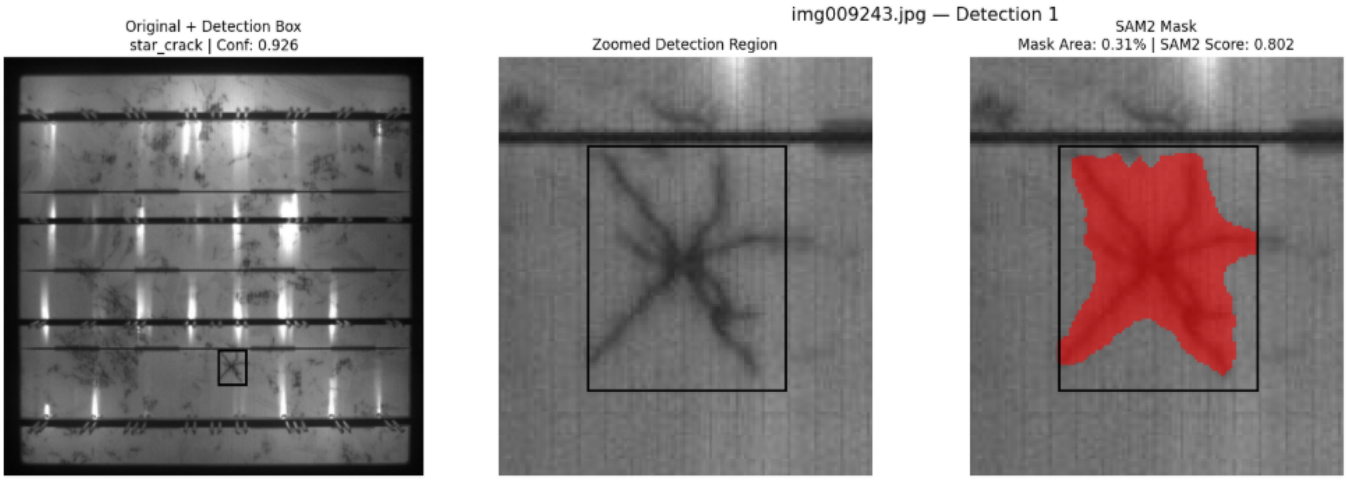


Fig. 8. SAM2-based segmentation of a star-crack defect.

E. Fault Severity Estimation

A heuristic fault severity estimation was calculated by using the relative bounding-box area and class-specific severity weights. As our dataset contains no ground-truth severity labels, we calculated an estimated severity level rather than a supervised severity prediction.

Table XI presents representative detections for the three severity levels. Low-severity defects occupied less than 1% of the image area, while medium-severity defects covered approximately 4–5%. The highest scores were observed for *short_circuit* detections, which produced bounding boxes covering approximately 89% of the image area. These large regions are influenced by the localization limitations of minority classes and therefore should not be interpreted as their actual physical defect area.

TABLE XI
REPRESENTATIVE FAULT SEVERITY ESTIMATION RESULTS

Severity	**Defect**	**Confidence**	**Area Ratio**	**Weighted Score**
Low	*star_crack*	0.958	0.8802%	0.011443
Medium	*crack*	0.648	4.9556%	0.059467
High	*short_circuit*	0.926	89.1863%	1.426980

Our results demonstrate that the proposed heuristic provides an interpretable indication of defect extent and relative severity.

F. Explainable AI (XAI)

TABLE XII
GRAD-CAM++ POINTING-GAME LOCALIZATION HIT-RATE (PVTV2-B0 + FPN + RETINANET, P3 LEVEL)

Class	**Hits**	**Total**	**Hit Rate**
crack	16	29	0.552
black_core	15	15	0.714
thick_line	13	13	0.650
horizontal_dislocation	03	21	0.143
short_circuit	02	13	0.154
vertical_dislocation	05	05	1.000
star_crack	03	06	0.500
Overall	**66**	**147**	**0.449**

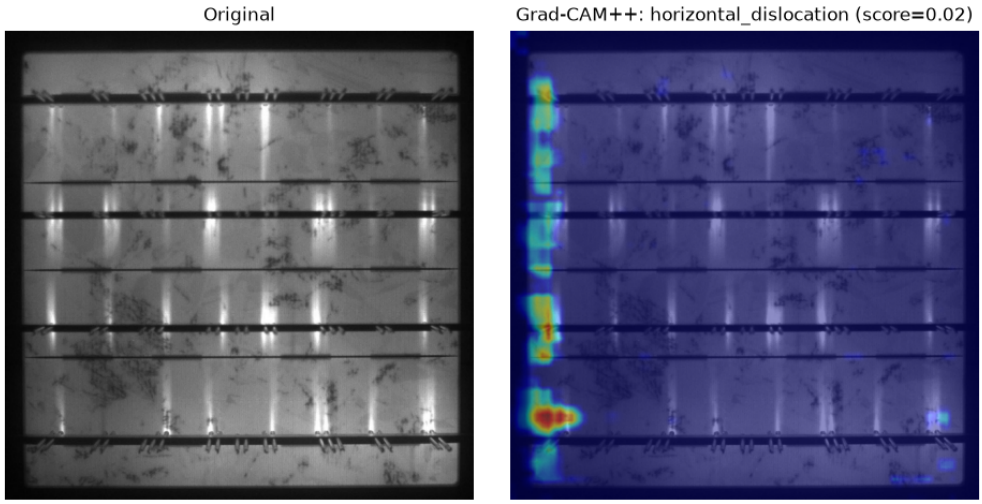


Fig. 9. Produced heatmap by GradCam++ for Horizontal Dislocation

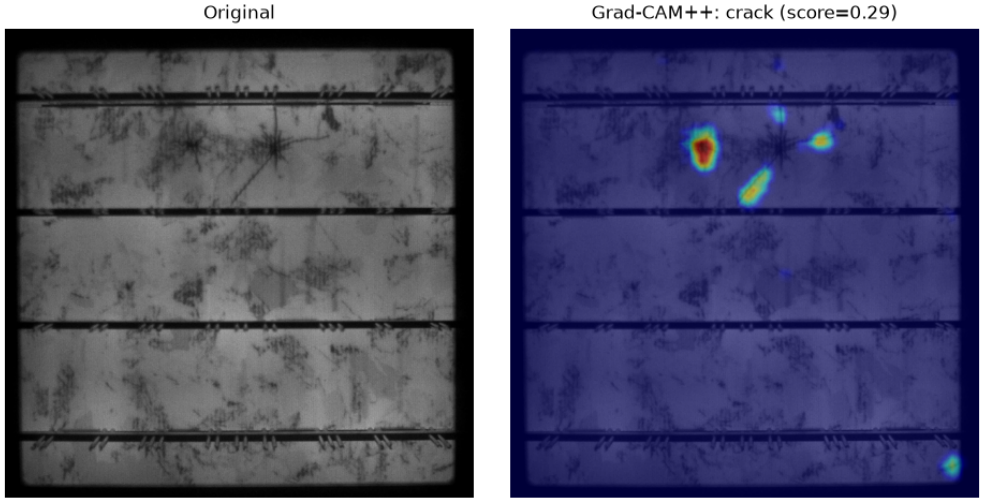


Fig. 10. Produced heatmap by GradCam++ for Crack

In Fig.4 and Fig.5, we have shown the produced hit map. During the testing, the XAI has achieved 44.9% heat rate overall, ranging from 100% for vertical_dislocation and 71.4% for black_core down to 15.4% for short_circuit and 14.3% for horizontal_dislocation, with the harder classes plausibly explained by defects that are thin or spread across several small regions rather than one compact area. Again in the Table XII, we have shown total hits and hit scores for each defective class using the Grad-Cam++.

TABLE XIII
COMPARISON WITH EXISTING METHODS FOR PV CELL DEFECT DETECTION

Ref.	**Dataset**	**Model**	**mAP@50**	**mAP@50-95**
[6]	[4]	MRA-YOLOv8	0.917	0.62
[7]	[4]	PV-YOLOv12n	0.91	0.58
[8]	EESD / PV Inspection Infrared	GDDS	–	–
Proposed	[4]	**PVT-MAAG**	**0.9085**	**0.6065**

Table XIII shows a thorough comparison of PV cell defect detection between the proposed hybrid PVT-MAAG model and several existing works. Although MRA-YOLOv8 attained slightly higher mAP@50 (0.917) and mAP@50-95 (0.62), the PVT-MAAG model delivers excellent performance, with mAP@50 (0.9085) and mAP@50-95 (0.6065). The PVT-MAAG model outperforms PV-YOLOv12n on the mAP@50-95 metric. Performance benchmarking based on the GDDS framework is difficult because the model has been trained and tested on different datasets (EESD/PV Inspection Infrared).

IV. Conclusion And Future Work

In this work, we have built and tested a complete PV cell defect detection system using EL images. In this, we have trained and compared six baseline detectors. Among them, 3 are CNN-based models, and the rest of the models are transformer-based models. After the comparison, we have extended the best performing models, PVTv2B0-RetinaNet, into a hybrid model, and we named it PVT-MAAG. We have turned PVTv2B0 into PVT-MAAG by adding AdaLora fine-tuning, bidirectional Mamba feature extraction, adaptive spatial feature fusion, and graph-based reasoning on top of the backbone. By applying all the novelties, our PVT-MAAG has performed the best and outperformed all the models in terms of performance tested on the test split, and gained the mAP@50 of 90.85% and mAP@50-95 of 60.65%. We have also distilled the PVT-MAAG by keeping the two lighter YOLO models as the student model. With this, our system is now deployable. We have also performed the Explainable AI on PVTv2B0-RetinaNet. As the XAI, we have used GradCam++. We have ensembled PVT-MAAG and the ResNet50-RetinaNet with the SAM2 segmentation and a rule-based fault severity estimator. By this, we have extended our developed system past raw detection towards an inspection that is interpretable and aware of defect severity.

We have several extensions for our future work.

We have only visually inspected the best ensembled models. This pipeline is fused with two detectors with the Weighted Box fusion. We then feed the results to the SAM-2 and severity estimator. We have now planned to perform a full qualitative comparison of the ensemble model against each detection model. We will update the ensemble if we find a more credible model for this.

In the future, we will work to make this system edge device compatible. For this, we will quantize our models or export the distilled student models to run directly on detection hardware. YOLOv8m and YOLOv11n are simpler, lighter models than the PVT-MAAG stack. During distillation, we have kept both of them as student models and the PVT-MAAG as the teacher. For this reason, we are considering the YOLOv8m and the YOLOv11n as the frontier models in this case.

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